Example 6.17

If the vectors $\vec{a},\vec{b},\vec{c}$ are coplanar, then prove that the vectors $\vec{a}+\vec{b},\vec{b}+\vec{c},\vec{c}+\vec{a}$ are also coplanar.

Solution

Since the vectors $\vec{a},\vec{b},\vec{c}$ are coplanar, we have $[\vec{a},\ \vec{b},\ \vec{c}]=0$. Using the properties of the scalar triple product, we get

$$\begin{aligned}[\vec{a}+\vec{b},\vec{b}+\vec{c},\vec{c}+\vec{a}] &= [\vec{a},\vec{b}+\vec{c},\vec{c}+\vec{a}]+[\vec{b},\vec{b}+\vec{c},\vec{c}+\vec{a}]\\ &= [\vec{a},\vec{b},\vec{c}+\vec{a}]+[\vec{a},\vec{c},\vec{c}+\vec{a}]+[\vec{b},\vec{b},\vec{c}+\vec{a}]+[\vec{b},\vec{c},\vec{c}+\vec{a}]\\ &= [\vec{a},\vec{b},\vec{c}]+[\vec{a},\vec{b},\vec{a}]+[\vec{a},\vec{c},\vec{c}]+[\vec{a},\vec{c},\vec{a}]+[\vec{b},\vec{b},\vec{c}]+[\vec{b},\vec{b},\vec{a}]+[\vec{b},\vec{c},\vec{c}]+[\vec{b},\vec{c},\vec{a}]\\ &= [\vec{a},\vec{b},\vec{c}]+[\vec{a},\vec{b},\vec{c}]=2[a,b,c]=0.\end{aligned}$$

Hence the vectors $\vec{a}+\vec{b},\ \vec{b}+\vec{c},\ \vec{c}+\vec{a}$ are coplanar.

Example 6.18

If $\vec{a},\vec{b},\vec{c}$ are three vectors, prove that $[\vec{a}+\vec{c},\ \vec{a}+\vec{b},\ \vec{a}+\vec{b}+\vec{c}]=[\vec{a},\vec{b},\vec{c}]$.

Solution

Using theorem 6.6, we get

$$[\vec{a}+\vec{c},\vec{a}+\vec{b},\ \vec{a}+\vec{b}+\vec{c}] = \begin{vmatrix}1 & 0 & 1\\ 1 & 1 & 0\\ 1 & 1 & 1\end{vmatrix}[\vec{a},\vec{b},\vec{c}]$$

$$= [\vec{a},\vec{b},\vec{c}].$$

EXERCISE 6.2

1. If $\vec{a}=\hat{i}-2\hat{j}+3\hat{k},\ \vec{b}=2\hat{i}+\hat{j}-2\hat{k},\ \vec{c}=3\hat{i}+2\hat{j}+\hat{k}$, find $\vec{a}\cdot(\vec{b}\times\vec{c})$.
2. Find the volume of the parallelepiped whose coterminous edges are represented by the vectors $-6\hat{i}+14\hat{j}+10\hat{k},\ 14\hat{i}-10\hat{j}-6\hat{k}$ and $2\hat{i}+4\hat{j}-2\hat{k}$.
3. The volume of the parallelepiped whose coterminus edges are $7\hat{i}+\lambda\hat{j}-3\hat{k},\ \hat{i}+2\hat{j}-\hat{k},$ $-3\hat{i}+7\hat{j}+5\hat{k}$ is 90 cubic units. Find the value of λ.
4. If $\vec{a},\vec{b},\vec{c}$ are three non-coplanar vectors represented by concurrent edges of a parallelepiped of volume 4 cubic units, find the value of $(\vec{a}+\vec{b})\cdot(\vec{b}\times\vec{c})+(\vec{b}+\vec{c})\cdot(\vec{c}\times\vec{a})+(\vec{c}+\vec{a})\cdot(\vec{a}\times\vec{b})$.
5. Find the altitude of a parallelepiped determined by the vectors $\vec{a}=-2\hat{i}+5\hat{j}+3\hat{k},\ \hat{b}=\hat{i}+3\hat{j}-2\hat{k}$ and $\vec{c}=-3\vec{i}+\vec{j}+4\vec{k}$ if the base is taken as the parallelogram determined by $\vec{b}$ and $\vec{c}$.
6. Determine whether the three vectors $2\hat{i}+3\hat{j}+\hat{k},\ \hat{i}-2\hat{j}+2\hat{k}$ and $3\hat{i}+\hat{j}+3\hat{k}$ are coplanar.
7. Let $\vec{a}=\hat{i}+\hat{j}+\hat{k},\ \vec{b}=\hat{i}$ and $\vec{c}=c_1\hat{i}+c_2\hat{j}+c_3\hat{k}$. If $c_1=1$ and $c_2=2$, find c_3 such that $\vec{a},\vec{b}$ and $\vec{c}$ are coplanar.

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